位图显示　　像素驱动位图

```
控制 时钟频率 = 10_000_000

PUB go() | i

  调试 ( 位图 MyBitmap 尺寸 32 16 点尺寸 8 LUT2 LONGS_2BIT)
  调试 (MyBitmap 追踪 14 LUT 颜色 白色 红色 蓝色 黄色 6)
  重复
    调试 (MyBitmap uhex_( 标志 [i++ & $1F]) dly(100))

数据

标志    长      已处理。
        长      ```zh%%0010101022222220```
        长      %%0010101020202020
        长      ```zh%%0010101022222220```
        长      %%0010101022020220
        长      ```zh%%0010101022222220```
        长      %%0010101020202020
        长      ```zh%%0010101022222220```
        长      %%0010101022020220
        长      ```zh%%0010101022222220```
        长      %%0010101020202020
        长      ```zh%%0010101022222220```
        长      %%0010101010101010
        长      %%0010101010101010
        长      %%0010101010101010
        长      %%0010101010101010
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        长      %%0010101010101010
        长      %%0010101010101010
        长      %%0010101010101010
        长      %%0010101010101010
        长      %%0010101010101010
        长      已处理。
        长      %%0000000000000000
                00000000000
                00000000000
        长      %%0000000000000000
```

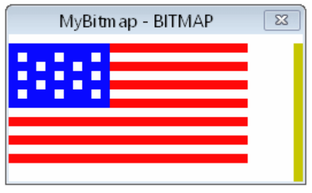

位图实例化	描述	默认
标题 ' 字符串 '	将窗口标题设置为 " 字符串 "。	无
位置 左上	设置窗口位置。	0, 0
尺寸 x_ 像素 y_ 像素	设置位图中像素的数量（ x 和 y 均为 1..2048 ）。	256, 256
DOTSIZE x_ 和 _y_ 尺寸 {y_ 尺寸 }	设置显示像素的尺寸 (1..256)。'DOTSIZE 16' 使最终显示中的每个像素为 16x16。	1, 1
lut1_to_rgb24	设置颜色模式。请参阅以下图片。	RGB24
LUT 色彩 rgb24 rgb24 ...	对于 LUT1..LUT8 颜色模式，请加载包含 RGB24 颜色的 LUT。使用 HEX_LONG_ARRAY_ 加载。	默认颜色 0..7
TRACE 0_to_15	设置像素加载方向以及每行填充后是否滚动。动画如下。	
RATE pixels_per_update	在每次显示更新之前设置像素数。「 RATE -1 」将速率设置为位图尺寸。	行大小
打包数据模式	启用紧凑数据模式。请参阅本节末尾的描述。	无
更新	设置更新模式。只有在接收到 " 更新 " 命令时，显示才会更新。	自动更新
位图馈送	**描述**	**默认**
lut1_to_rgb24	更改颜色模式。	RGB24
LUT 色彩 rgb24 rgb24 ...	对于 LUT1..LUT8 颜色模式，请加载包含 rgb24 颜色的 LUT。使用 HEX_LONG_ARRAY_ 加载颜色。	默认颜色 0..7
TRACE 0_to_15	改变像素加载到位图的方向。将速率设置为行大小。	0
RATE pixels_per_update	在每次显示更新之前设置像素数。「 RATE -1 」将速率设置为位图尺寸。	
将 x 位置设置为 y 位置	设置当前像素加载位置。清空 TRACE 的第 3 位，从而取消滚动模式。	
滚动 x 滚动 y 滚动	滚动位图若干像素。负值 / 正值确定方向，无。	
清空	清空位图中的值。	
更新	使用当前位图更新窗口。在 UPDATE 模式下使用。	
保存 {WINDOW} ' 文件名 '	保存整个窗口或仅保存 1x 比例的位图文件（ .bmp ）的位图文件。	
关闭	关闭窗口。	

BITMAP Display　　Pixel-driven bitmap

```
CON _clkfreq = 10_000_000

PUB go() | i

  debug(`bitmap MyBitmap SIZE 32 16 DOTSIZE 8 LUT2 LONGS_2BIT)
  debug(`MyBitmap TRACE 14 LUTCOLORS WHITE RED BLUE YELLOW 6)
  repeat
    debug(`MyBitmap `uhex_(flag[i++ & $1F]) `dly(100))

DAT

flag    long    %%3333333333333330
        long    %%0010101022222220
        long    %%0010101020202020
        long    %%0010101022222220
        long    %%0010101022020220
        long    %%0010101022222220
        long    %%0010101020202020
        long    %%0010101022222220
        long    %%0010101022020220
        long    %%0010101022222220
        long    %%0010101020202020
        long    %%0010101022222220
        long    %%0010101010101010
        long    %%0010101010101010
        long    %%0010101010101010
        long    %%0010101010101010
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        long    %%0010101010101010
        long    %%0010101010101010
        long    %%0010101010101010
        long    %%0010101010101010
        long    %%0010101010101010
        long    %%0010101010101010
        long    %%0010101010101010
        long    %%0000000000000000
        long    %%0000000000000000
        long    %%0000000000000000
        long    %%0000000000000000
        long    %%0000000000000000
```
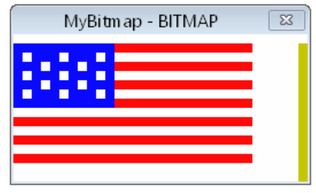
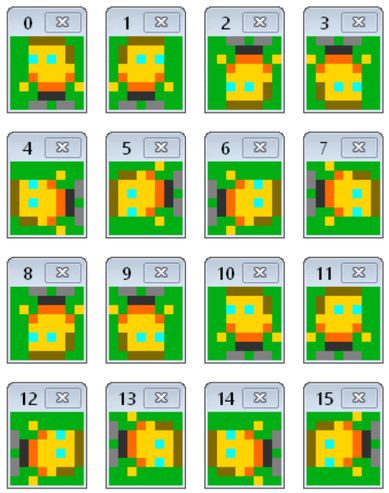



BITMAP Instantiation	Description	Default
TITLE 'string'	Set the window caption to 'string'.	<none>
POS left top	Set the window position.	0, 0
SIZE x_pixels y_pixels	Set the number of pixels in the bitmap (1..2048 for both x and y).	256, 256
DOTSIZE x_and_y_size {y_size}	Set the size of the displayed pixels (1..256). 'DOTSIZE 16' makes each pixel 16x16 on the final display.	1, 1
lut1_to_rgb24	Set the color mode. See images below.	RGB24
LUTCOLORS rgb24 rgb24 ...	For LUT1..LUT8 color modes, load the LUT with RGB24 colors. Use HEX_LONG_ARRAY_ to load.	default colors 0..7
TRACE 0_to_15	Set the pixel loading direction and whether to scroll after each line is filled. See animation below.	0
RATE pixels_per_update	Set the number of pixels before each display update. 'RATE -1' sets the rate to the bitmap size.	line size
packed_data_mode	Enable packed-data mode. See description at end of this section.	<none>
UPDATE	Set UPDATE mode. The display will only be updated when fed an 'UPDATE' command.	automatic update
BITMAP Feeding	**Description**	**Default**
lut1_to_rgb24	Change the color mode.	RGB24
LUTCOLORS rgb24 rgb24 ...	For LUT1..LUT8 color modes, load the LUT with rgb24 colors. Use HEX_LONG_ARRAY_ to load colors.	default colors 0..7
TRACE 0_to_15	Change the direction in which pixels are loaded into the bitmap. Sets the rate to the line size.	0
RATE pixels_per_update	Set the number of pixels before each display update. 'RATE -1' sets the rate to the bitmap size.	
SET x_position {y_position}	Set the current pixel-loading position. Cancels scroll mode by clearing bit 3 of TRACE.	
SCROLL x_scroll y_scroll	Scroll the bitmap by some number of pixels. Negative/positive values determine the direction, 0 = none.	
CLEAR	Clear the bitmap to zero-value pixels.	
UPDATE	Update the window with the current bitmap. Used in UPDATE mode.	
SAVE {WINDOW} 'filename'	Save a bitmap file (.bmp) of either the entire window or just the bitmap at 1x scale.	
CLOSE	Close the window.	

TRACE modes

Rate is set to 1 so that each pixel can be seen as it's loaded.

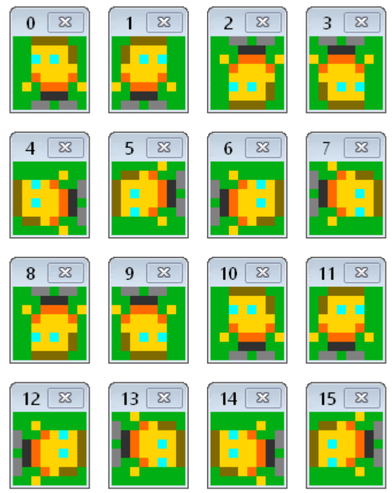


颜色模式	位像素	描述	意图
LUT1	1	像素索引 LUT 颜色 0..1	内存效率高 2 色调色板图形
LUT2	2	像素索引 LUT 颜色 0..3	Memory-efficient 4-color-palette graphics
LUT4	4	像素索引 LUT 颜色 0..15	高效的 16 颜色 调色板 图形
LUT8	8	像素索引 LUT 颜色 0..255	高效的 256 颜色 调色板 图形。
LUMA8	8	从黑色到颜色 *	仪器，其中亮度表示水平
LUMA8W	8	从白色到颜色 *	仪器，其中饱和度表示电平
LUMA8X	8	从黑色到颜色 * 到白色	仪器，亮度指示水平，峰值在白色
HSV8	8	从黑色到颜色：%HHHHSSSS	16 色调，16 亮度级别
HSV8W	8	从白色到颜色：%HHHHSSSS	16 种色调，包含 16 种饱和度级别，源自白色
HSV8X	8	从黑色到颜色到白色：%HHHHSSSS	16 种色调，包含 16 种亮度级别，峰值在白色。
RGBI8	8	从黑色到颜色：%RGBIIIII	8 种基本颜色，包含 32 个亮度级别
RGBI8W	8	白色到颜色：%RGBIIIII	8 种基本颜色，有 32 个饱和度级别，从白色而来。
RGBI8X	8	从黑色到颜色到白色：%RGBIIIII	8 种基本颜色，具有 32 级亮度，峰值在白色。
RGB8	8	%RRRGGGBB	字节级 RGB，包含 8 级红色、 8 级绿色和 4 级蓝色
HSV16	16	从黑色到颜色：%HHHHHHHH_SSSSSSSS	256 种色调和 256 级亮度
HSV16W	16	从白色到颜色：%HHHHHHHH_SSSSSSSS	256 种色调，包含 256 种饱和度级别，源自白色
HSV16X	16	从黑色到颜色到白色：%HHHHHHHH_SSSSSSSS	256 种色调，256 级亮度，峰值在白色
RGB16	16	%RRRRRGGG_GGGBBBBB	32 红色级别、 64 绿色级别和 32 蓝色级别的字元级 RGB
RGB24	24	%RRRRRRRR_GGGGGGGG_BBBBBBBB	全 RGB 颜色通道，红色、绿色和蓝色各 256 级

颜色 i s ORANGE / BLUE / GREEN / CYAN / RED / MAGENTA / YELLOW / GREY.

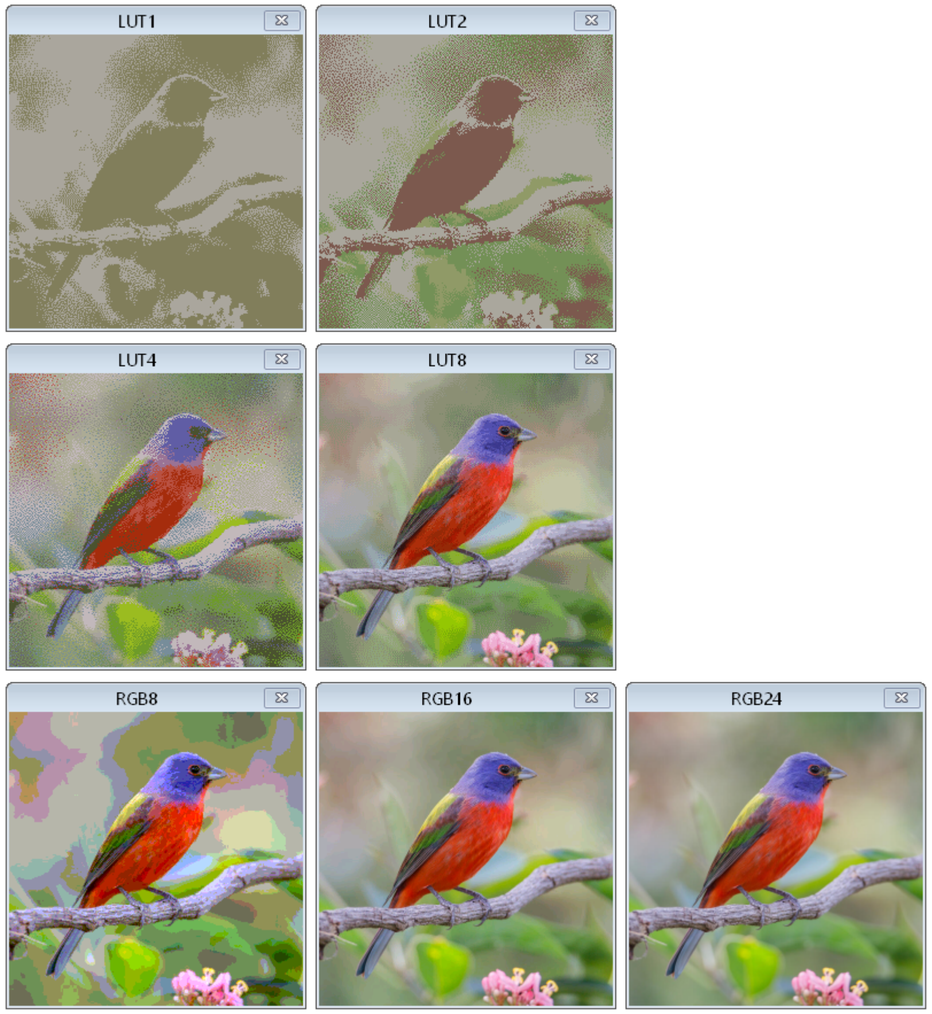
TRACE modes

Rate is set to 1 so that each pixel can be seen as it's loaded.

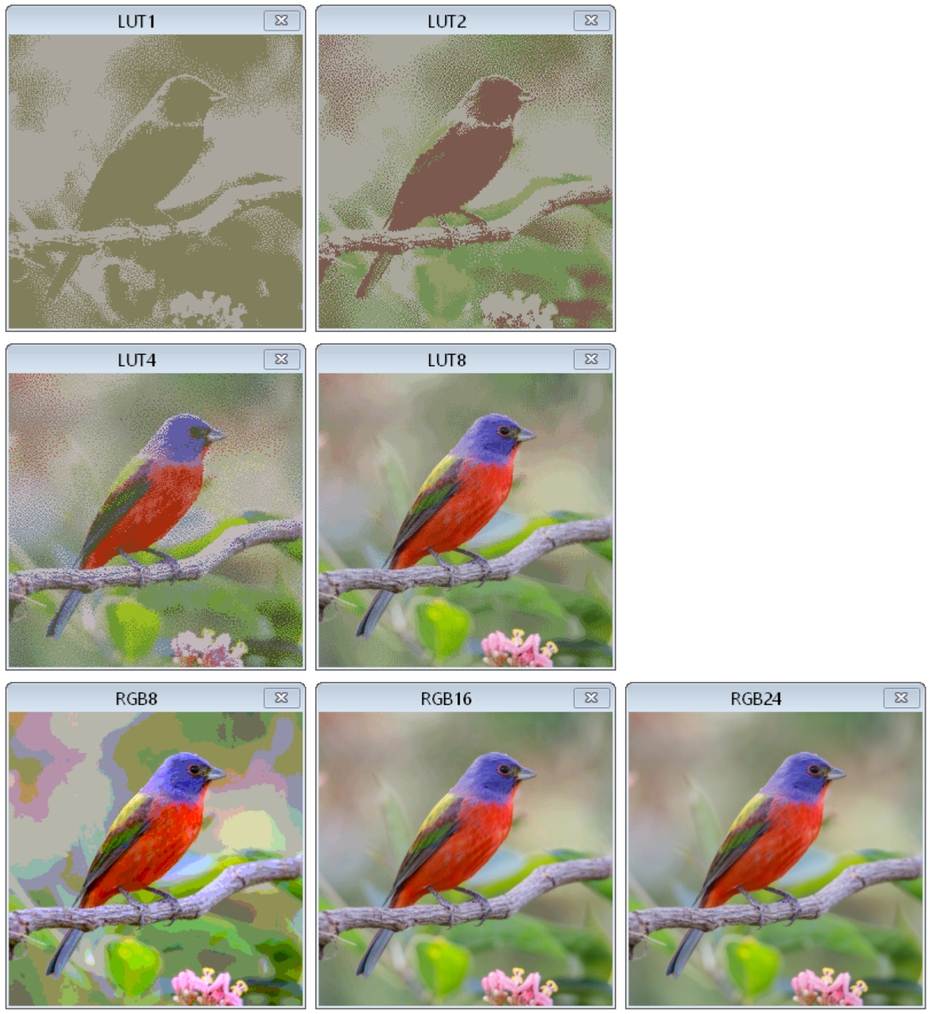


Color Mode	Bits/ Pixel	Description	Intention
LUT1	1	Pixel indexes LUT colors 0..1	Memory-efficient 2-color-palette graphics
LUT2	2	Pixel indexes LUT colors 0..3	Memory-efficient 4-color-palette graphics
LUT4	4	Pixel indexes LUT colors 0..15	Memory-efficient 16-color-palette graphics
LUT8	8	Pixel indexes LUT colors 0..255	Memory-efficient 256-color-palette graphics.
LUMA8	8	From black to color *	Instrumentation where luminance indicates level
LUMA8W	8	From white to color *	Instrumentation where saturation indicates level
LUMA8X	8	From black to color * to white	Instrumentation where luminance indicates level, peaking in white
HSV8	8	From black to color: %HHHHSSSS	16 hues with 16 luminance levels
HSV8W	8	From white to color: %HHHHSSSS	16 hues with 16 saturation levels, coming from white
HSV8X	8	From black to color to white: %HHHHSSSS	16 hues with 16 luminance levels, peaking in white
RGBI8	8	From black to color: %RGBIIIII	8 basic colors with 32 luminance levels
RGBI8W	8	From white to color: %RGBIIIII	8 basic colors with 32 saturation levels, coming from white
RGBI8X	8	From black to color to white: %RGBIIIII	8 basic colors with 32 luminance levels, peaking in white
RGB8	8	%RRRGGGBB	Byte-level RGB with 8 red, 8 green, and 4 blue levels
HSV16	16	From black to color: %HHHHHHHH_SSSSSSSS	256 hues with 256 luminance levels
HSV16W	16	From white to color: %HHHHHHHH_SSSSSSSS	256 hues with 256 saturation levels, coming from white
HSV16X	16	From black to color to white: %HHHHHHHH_SSSSSSSS	256 hues with 256 luminance levels, peaking in white
RGB16	16	%RRRRRGGG_GGGBBBBB	Word-level RGB with 32 red levels, 64 green levels, and 32 blue levels
RGB24	24	%RRRRRRRR_GGGGGGGG_BBBBBBBB	Full RGB with 256 levels for red, green, and blue

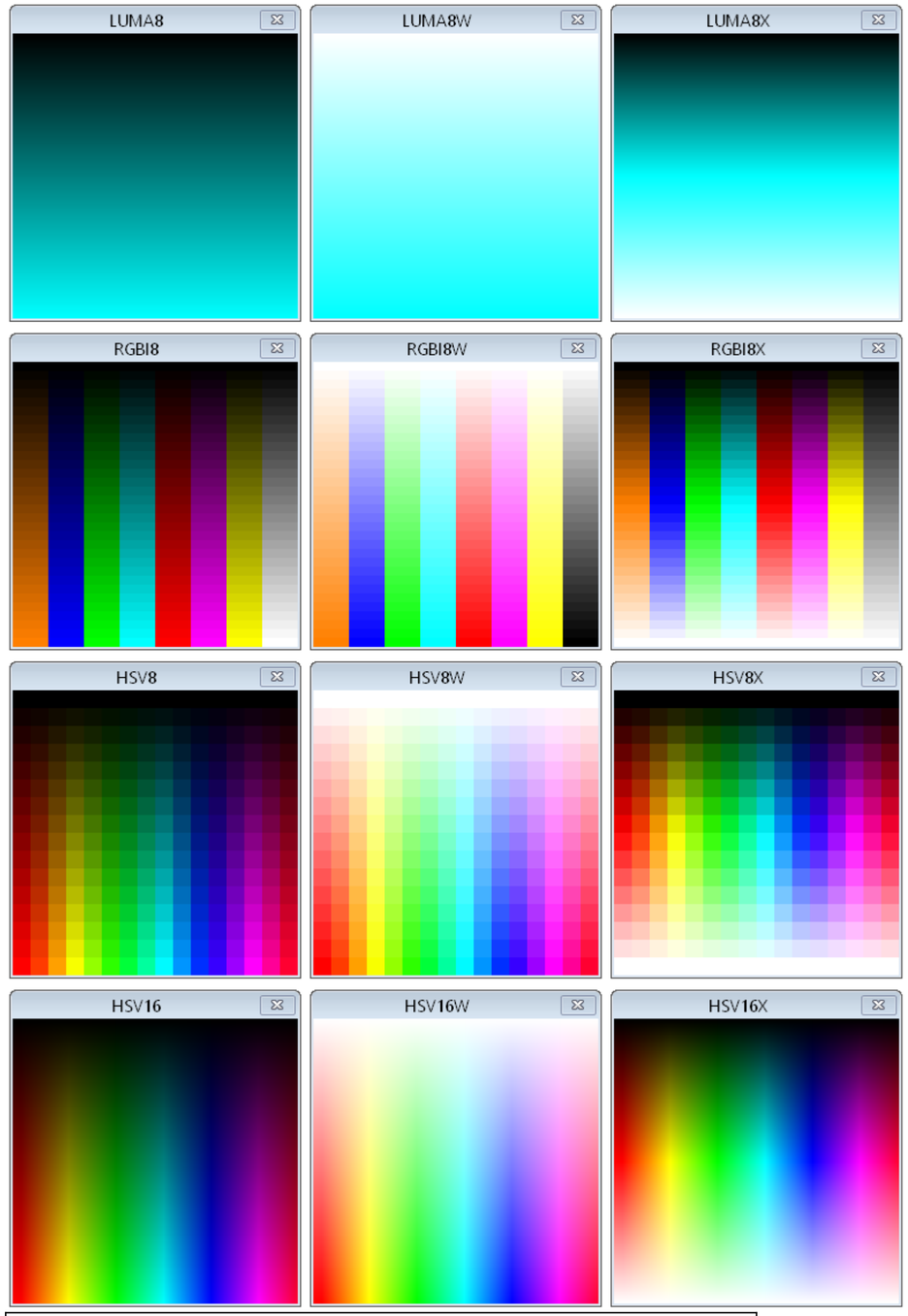
* Color is ORANGE / BLUE / GREEN / CYAN / RED / MAGENTA / YELLOW / GREY.




```
CON _clkfreq = 100_000_000

PUB go() | i
  debug(`bitmap a title 'LUT1'  pos 100 100 trace 2 lut1 longs_1bit alt)
  debug(`bitmap b title 'LUT2'  pos 370 100 trace 2 lut2 longs_2bit alt)
  debug(`bitmap c title 'LUT4'  pos 100 395 trace 2 lut4 longs_4bit alt)
  debug(`bitmap d title 'LUT8'  pos 370 395 trace 2 lut8 longs_8bit)
  debug(`bitmap e title 'RGB8'  pos 100 690 trace 2 rgb8)
  debug(`bitmap f title 'RGB16' pos 370 690 trace 2 rgb16)
  debug(`bitmap g title 'RGB24' pos 640 690 trace 2 rgb24)
  waitms(1000)

  showbmp("a", @image1, $8A,   2,  $800)    'send LUT1 image
  showbmp("b", @image2, $36,   4, $1000)    'send LUT2 image
  showbmp("c", @image3, $8A,  16, $2000)    'send LUT4 image
  showbmp("d", @image4, $36, 256, $4000)    'send LUT8 image

  i := @image5 + $36  'send RGB8/RGB16/RGB24 images from the same 24-bpp file
  repeat $10000
    debug(`e `uhex_(byte[i+0] >> 6 + byte[i+1] >> 5 << 2 + byte[i+2] >> 5 << 5 ))
    debug(`f `uhex_(byte[i+0] >> 3 + byte[i+1] >> 2 << 5 + byte[i+2] >> 3 << 11))
    debug(`g `uhex_(byte[i+0]      + byte[i+1] << 8      + byte[i+2] << 16     ))
    i += 3

PRI showbmp(letter, image_address, lut_offset, lut_size, image_longs) | i
  image_address += lut_offset
  debug(``#(letter) lutcolors `uhex_long_array_(image_address, lut_size))
  image_address += lut_size << 2 - 4
  repeat image_longs
    debug(``#(letter) `uhex_(long[image_address += 4]))

DAT
image1 file "bird_lut1.bmp"
image2 file "bird_lut2.bmp"
image3 file "bird_lut4.bmp"
image4 file "bird_lut8.bmp"
image5 file "bird_rgb24.bmp"
```


```
CON _clkfreq = 100_000_000

PUB go() | i
  debug(`bitmap a title 'LUT1'  pos 100 100 trace 2 lut1 longs_1bit alt)
  debug(`bitmap b title 'LUT2'  pos 370 100 trace 2 lut2 longs_2bit alt)
  debug(`bitmap c title 'LUT4'  pos 100 395 trace 2 lut4 longs_4bit alt)
  debug(`bitmap d title 'LUT8'  pos 370 395 trace 2 lut8 longs_8bit)
  debug(`bitmap e title 'RGB8'  pos 100 690 trace 2 rgb8)
  debug(`bitmap f title 'RGB16' pos 370 690 trace 2 rgb16)
  debug(`bitmap g title 'RGB24' pos 640 690 trace 2 rgb24)
  waitms(1000)

  showbmp("a", @image1, $8A,   2,  $800)    'send LUT1 image
  showbmp("b", @image2, $36,   4, $1000)    'send LUT2 image
  showbmp("c", @image3, $8A,  16, $2000)    'send LUT4 image
  showbmp("d", @image4, $36, 256, $4000)    'send LUT8 image

  i := @image5 + $36  'send RGB8/RGB16/RGB24 images from the same 24-bpp file
  repeat $10000
    debug(`e `uhex_(byte[i+0] >> 6 + byte[i+1] >> 5 << 2 + byte[i+2] >> 5 << 5 ))
    debug(`f `uhex_(byte[i+0] >> 3 + byte[i+1] >> 2 << 5 + byte[i+2] >> 3 << 11))
    debug(`g `uhex_(byte[i+0]      + byte[i+1] << 8      + byte[i+2] << 16     ))
    i += 3

PRI showbmp(letter, image_address, lut_offset, lut_size, image_longs) | i
  image_address += lut_offset
  debug(``#(letter) lutcolors `uhex_long_array_(image_address, lut_size))
  image_address += lut_size << 2 - 4
  repeat image_longs
    debug(``#(letter) `uhex_(long[image_address += 4]))

DAT
image1 file "bird_lut1.bmp"
image2 file "bird_lut2.bmp"
image3 file "bird_lut4.bmp"
image4 file "bird_lut8.bmp"
image5 file "bird_rgb24.bmp"
```




```
CON _clkfreq = 100_000_000

PUB go() | i
  debug(`bitmap a title 'LUMA8'  pos 100 100 size 1 256 dotsize 256 1 luma8  cyan)
  debug(`bitmap b title 'LUMA8W' pos 370 100 size 1 256 dotsize 256 1 luma8w cyan)
  debug(`bitmap c title 'LUMA8X' pos 640 100 size 1 256 dotsize 256 1 luma8x cyan)
  debug(`bitmap d title 'RGBI8'  pos 100 395 size 8 32 dotsize 32 8 trace 4 rgbi8)
  debug(`bitmap e title 'RGBI8W' pos 370 395 size 8 32 dotsize 32 8 trace 4 rgbi8w)
  debug(`bitmap f title 'RGBI8X' pos 640 395 size 8 32 dotsize 32 8 trace 4 rgbi8x)
  debug(`bitmap g title 'HSV8'   pos 100 690 size 16 16 trace 4 dotsize 16 hsv8)
  debug(`bitmap h title 'HSV8W'  pos 370 690 size 16 16 trace 4 dotsize 16 hsv8w)
  debug(`bitmap i title 'HSV8X'  pos 640 690 size 16 16 trace 4 dotsize 16 hsv8x)
  debug(`bitmap j title 'HSV16'  pos 100 985 size 256 256 trace 4 hsv16)
  debug(`bitmap k title 'HSV16W' pos 370 985 size 256 256 trace 4 hsv16w)
  debug(`bitmap l title 'HSV16X' pos 640 985 size 256 256 trace 4 hsv16x)
  waitms(1000)
  repeat i from 0 to 255                    'feed 8-bit displays
    debug(`a b c d e f g h i `uhex_(i))
  repeat i from 0 to 65535                  'feed 16-bit displays
    debug(`j k l `uhex_(i))
```

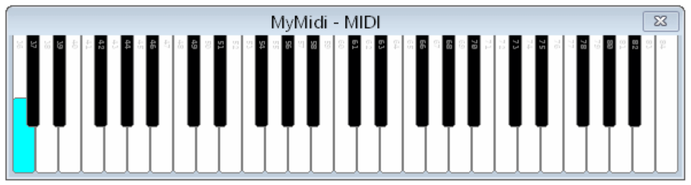

CON _clkfreq = 100_000_000

PUB go() | i
  debug(`bitmap a title 'LUMA8' pos 100 100 size 1 256 dotsize 256 1 luma8 cyan)
  debug(`bitmap b title 'LUMA8W' pos 370 100 size 1 256 dotsize 256 1 luma8w cyan)
  debug(`bitmap c title 'LUMA8X' pos 640 100 size 1 256 dotsize 256 1 luma8x cyan)
  debug(`bitmap d title 'RGBI8' pos 100 395 size 8 32 dotsize 32 8 trace 4 rgbi8)
  debug(`bitmap e title 'RGBI8W' pos 370 395 size 8 32 dotsize 32 8 trace 4 rgbi8w)
  debug(`bitmap f title 'RGBI8X' pos 640 395 size 8 32 dotsize 32 8 trace 4 rgbi8x)
  debug(`bitmap g title 'HSV8'   pos 100 690 size 16 16 trace 4 dotsize 16 hsv8)
  debug(`bitmap h title 'HSV8W' pos 370 690 size 16 16 trace 4 dotsize 16 hsv8w)
  debug(`bitmap i title 'HSV8X' pos 640 690 size 16 16 trace 4 dotsize 16 hsv8x)
  debug(`bitmap j title 'HSV16' pos 100 985 size 256 256 trace 4 hsv16)
  debug(`bitmap k title 'HSV16W' pos 370 985 size 256 256 trace 4 hsv16w)
  debug(`bitmap l title 'HSV16X' pos 640 985 size 256 256 trace 4 hsv16x)
  waitms(1000)
  repeat i from 0 to 255   'feed 8-bit displays
    debug(`a b c d e f g h i `uhex_(i))
  repeat i from 0 to 65535   'feed 16-bit displays
    debug(`j k l `uhex_(i))

MIDI 显示

用于查看音符上 / 下状态及音量的 MIDI 键盘

```
控制 CON _clkfreq = 10_000_000 PUB go() | i 调试 (`midi MyMidi 尺寸 3 范围 36 84) 重复 i 从 36 到 84 重复
调试 (`MyMidi $90 `(i, getrnd() & $7F)) 等待 150 调试
(`MyMidi $80 `(i, 0))
```

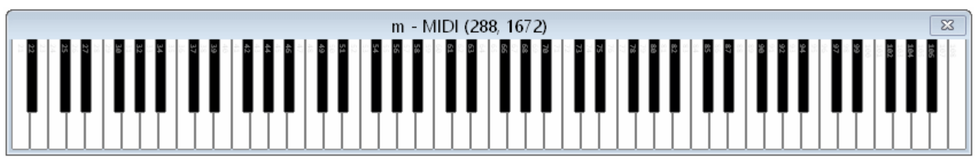


MIDI 初始化	描述	默认
标题 ' 字符串 '	将窗口标题设置为 " 字符串 "。	无
位置 左上	设置窗口位置。	0, 0
尺寸 键盘尺寸	设置 MIDI 键盘显示尺寸 (1..50)。	4
范围 起始键 结束键	设置起始和结束的 MIDI 键号 (0..127)。	21, 108 (88 个键)
通道 通道号	将 MIDI 通道号设置为观察 (0..15)。	0
白色白键 黑键	为白色和黑色键设置 "ON" 颜色。	青色，品红色
MIDI 输入	**描述**	**默认**
字节	如果 $90 + 通道，则为音符开启模式；否则，为音符关闭模式。 如果进入音符开启模式，则接收一个键（$00..$7F），然后是它的音量（$00..$7F），更新显示。 如果进入音符关闭模式，则接收一个键 ($00..$7F)，并更新其音量 ($00..$7F) 显示。	
清空	清空所有笔记。	
保存 窗口 ' 文件名 '	保存整个窗口或仅显示区域的位图文件（.bmp）。	
关闭	关闭窗口。	

* Color is BLACK / WHITE or ORANGE / BLUE / GREEN / CYAN / RED / MAGENTA / YELLOW / GREY followed by an optional 0..15 for brightness (default is 8).

Here is a PASM program which receives MIDI serial on P16 and sends it to the MIDI display:

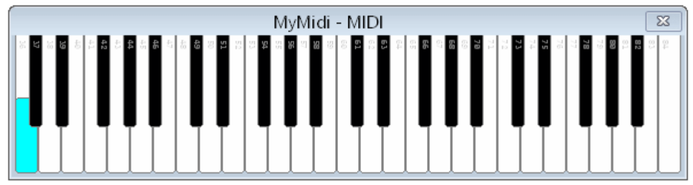
```
控制    时钟频率        = 10_000_000
        输入引脚        = 16
数据    组织

        汇编时钟

        调试 (midi m 大小 2)

        写入引脚 输入引脚
        写入输出引脚    # ( 时钟频率 /31250) << 16 + 8-1, 输入引脚
        驱动程序        输入引脚
.wait testp #rxpin wc
        如果条件跳转    #.wait

        rdpiin  x,#rxpin
        shr     x,#32-8

        调试 ("` m ", x)

        跳转    #.wait

x       复位    1
```



MIDI Display

MIDI keyboard for viewing note-on/off status with velocity

```
CON _clkfreq = 10_000_000

PUB go() | i

  debug(`midi MyMidi size 3 range 36 84)
  repeat
    repeat i from 36 to 84
      debug(`MyMidi $90 `(i, getrnd() & $7F))
      waitms(150)
      debug(`MyMidi $80 `(i, 0))
```

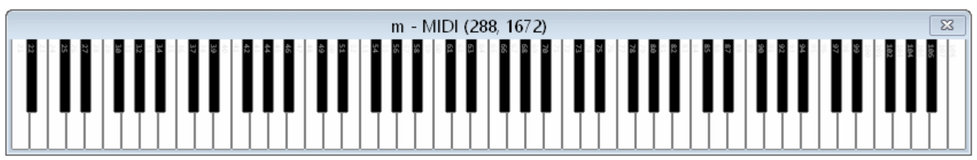


MIDI Instantiation	Description	Default
`TITLE 'string'`	Set the window caption to 'string'.	<none>
`POS left top`	Set the window position.	0, 0
`SIZE keyboard_size`	Set the size of the MIDI keyboard display (1..50).	4
`RANGE first_key last_key`	Set the first and last MIDI key numbers (0..127).	21, 108 (88 keys)
`CHANNEL channel_number`	Set the MIDI channel number to observe (0..15).	0
`COLOR white_key black_key`	Set the 'ON' colors for white and black keys. *	CYAN, MAGENTA
MIDI Feeding	**Description**	**Default**
`byte`	If $90 + channel then NOTE_ON mode, else if $80 + channel then NOTE_OFF mode. If NOTE_ON mode then receive a key ($00..$7F) and then its velocity ($00..$7F), update display. If NOTE_OFF mode then receive a key ($00..$7F) and then its velocity ($00..$7F), update display.	
`CLEAR`	Clear all notes.	
`SAVE {WINDOW} 'filename'`	Save a bitmap file (.bmp) of either the entire window or just the display area.	
`CLOSE`	Close the window.	

* Color is BLACK / WHITE or ORANGE / BLUE / GREEN / CYAN / RED / MAGENTA / YELLOW / GREY followed by an optional 0..15 for brightness (default is 8).

Here is a PASM program which receives MIDI serial on P16 and sends it to the MIDI display:

```
CON     _clkfreq        = 10_000_000
        rxpin           = 16

DAT     org

        asmclk

        debug   (`midi m size 2)

        wrpin   #%11111_0,#rxpin
        wxpin   ##(clkfreq_/31250) << 16 + 8-1, #rxpin
        drvl    #rxpin

.wait   testp   #rxpin wc
  if_nc jmp     #.wait

        rdpin   x,#rxpin
        shr     x,#32-8

        debug   ("`m ", uhex_byte_(x))

        jmp     #.wait

x       res     1
```



紧凑数据模式

紧凑数据模式用于高效地传输小于字节的数据类型，通过主机侧对其接收到的字节、字节或长字节进行解包来实现。此外，字节可以在字节和长字节之间发送，并且字节可以在长字节之间发送，以提高效率。

为了建立紧凑数据操作，您必须指定以下任一模式，后跟可选的 “ALT” 和 “SIGNED” 关键词：

紧凑模式 ALT SIGNED

ALT 关键字会导致发送的每个字节内的位、双位或 nibble 在主机侧重新排序。这简化了原始数据在与您使用的调试显示相比时，其位字段顺序不正确的情况。这最有可能适用于以标准格式组成的位图数据。

SIGNED 关键字会导致所有未解包的数据值在主机侧进行符号扩展。

紧凑数据模式	描述	最终值	最终值 如果 SIGNED
长	接收到的每个值都翻译成 32 个单独的 1 位值，从接收值的最低有效位开始。	0..1	-1..0
长	接收到的每个值都翻译成 16 个单独的 2 位值，从接收到的值的最低有效位开始。	0..3	-2..1
长	接收到的每个值都翻译成 8 个单独的 4 位值，从接收值中的最低有效位开始。	0..15	-8..7
长	接收到的每个值都翻译成 4 个单独的 8 位值，从接收值的最低有效位开始。	0..255	-128..127
16 位长整型	接收到的每个值都翻译成 2 个独立的 16 位值，从接收值的最低有效位开始。	0..65,535	-32768..32767
1 位字型	接收到的每个值都翻译成 16 个单独的 1 位值，从接收值的最低有效位开始。	0..1	-1..0
2 位字型	接收到的每个值都翻译成 8 个单独的 2 位值，从接收值中的最低有效位开始。	0..3	-2..1
4 位元数据模式	接收到的每个值都翻译成 4 个单独的 4 位值，从接收值的最低有效位开始。	0..15	-8..7
8 位	接收到的每个值都翻译成 2 个单独的 8 位值，从接收值的最低有效位开始。	0..255	-128..127
字节 _1 位	接收到的每个值都翻译成 8 个单独的 1 位值，从接收值的最低有效位开始。	0..1	-1..0
2 位字节	接收到的每个值都翻译成 4 个单独的 2 位值，从接收到的值的最低有效位开始。	0..3	-2..1
4 位字节	接收到的每个值都翻译成 2 个单独的 4 位值，从接收值的最低有效位开始。	0..15	-8..7

Packed-Data Modes

Packed-data modes are used to efficiently convey sub-byte data types, by having the host side unpack them from bytes, words, or longs it receives. As well, bytes can be sent within words and longs, and words can be sent within longs for some efficiency improvement.

To establish packed-data operation, you must specify one of the modes listed below, followed by optional 'ALT' and 'SIGNED' keywords:

```
packed_mode {ALT} {SIGNED}
```

The **ALT** keyword will cause bits, double-bits, or nibbles, within each byte sent, to be reordered on the host side, within each byte. This simplifies cases where the raw data you are sending has its bitfields out-of-order with respect to the DEBUG display you are using. This is most-likely to be needed for bitmap data that was composed in standard formats.

The **SIGNED** keyword will cause all unpacked data values to be sign-extended on the host side.

Packed-Data Modes	Descriptions	Final Values	Final Values if SIGNED
LONGS_1BIT	Each value received is translated into 32 separate 1-bit values, starting from the LSB of the received value.	0..1	-1..0
LONGS_2BIT	Each value received is translated into 16 separate 2-bit values, starting from the LSBs of the received value.	0..3	-2..1
LONGS_4BIT	Each value received is translated into 8 separate 4-bit values, starting from the LSBs of the received value.	0..15	-8..7
LONGS_8BIT	Each value received is translated into 4 separate 8-bit values, starting from the LSBs of the received value.	0..255	-128..127
LONGS_16BIT	Each value received is translated into 2 separate 16-bit values, starting from the LSBs of the received value.	0..65,535	-32,768..32,767
WORDS_1BIT	Each value received is translated into 16 separate 1-bit values, starting from the LSB of the received value.	0..1	-1..0
WORDS_2BIT	Each value received is translated into 8 separate 2-bit values, starting from the LSBs of the received value.	0..3	-2..1
WORDS_4BIT	Each value received is translated into 4 separate 4-bit values, starting from the LSBs of the received value.	0..15	-8..7
WORDS_8BIT	Each value received is translated into 2 separate 8-bit values, starting from the LSBs of the received value.	0..255	-128..127
BYTES_1BIT	Each value received is translated into 8 separate 1-bit values, starting from the LSB of the received value.	0..1	-1..0
BYTES_2BIT	Each value received is translated into 4 separate 2-bit values, starting from the LSBs of the received value.	0..3	-2..1
BYTES_4BIT	Each value received is translated into 2 separate 4-bit values, starting from the LSBs of the received value.	0..15	-8..7

智能别针配置的内置符号

智能别针 符号 值	符号名称	详情
A 输入 极性	**打包数据模式**	
%0000_0000_000_000000000000_00_00000_0	真 (默认)	真 A 输入
%1000_0000_000_000000000000_00_00000_0	反转 A	反转 A 输入
输入选择	**打包数据模式**	
%0000_0000_000_000000000000_00_00000_0	A 的本地密码 (默认)	选择 A 输入的本地针脚
%0001_0000_000_000000000000_00_00000_0	A 的密码 +1	选择 A 输入中的针脚 +1
%0010_0000_000_000000000000_00_00000_0	A 的密码 +2	选择 A 输入中的针脚 +2
%0011_0000_000_000000000000_00_00000_0	A 的密码 +3	选择 A 输入中的针脚 +3
%0100_0000_000_000000000000_00_00000_0	P_ 输出 _A	选择 A 输入的输出位
%0101_0000_000_000000000000_00_00000_0	P_MINUS3_A	选择 引脚 -3 用于 A 输入
%0110_0000_000_000000000000_00_00000_0	P_MINUS2_A	选择 引脚 -2 用于 A 输入
%0111_0000_000_000000000000_00_00000_0	P_MINUS1_A	选择 引脚 -1 用于 A 输入
B 输入 极性	**打包数据模式**	
%0000_0000_000_000000000 0000_00_00000_0	真 (默认)	真 B 输入
%0000_1000_000_000000000000_00_00000_0	反转 B	反转 B 输入
B 输入选择	**打包数据模式**	
%0000_0000_000_000000000000_00_00000_0	P_LOCAL_B (默认)	为 B 输入选择本地 PIN
%0000_0001_000_000000000000_00_00000_0	P_PLU+1_B	选择 B 输入的引脚 +1
%0000_0010_000_000000000000_00_00000_0	P_PLUS2_B	选择 B 输入的引脚 +2
%0000_0011_000_000000000000_00_00000_0	P_PLUS3_B	选择 B 输入的引脚 +3
%0000_0100_000_000000000000_00_00000_0	B_ 输出	选择 B 输入的输出位
%0000_0101_000_000000000000_00_00000_0	P_MINUS3_B	选择 引脚 -3 以获得 B 输入
%0000_0110_000_000000000000_00_00000_0	P_MINUS2_B	选择 引脚 -2 以获得 B 输入
%0000_0111_000_000000000000_00_00000_0	P_MINUS1_B	选择 引脚 -1 以获得 B 输入
A, B 输入 逻辑	**打包数据模式**	
%0000_0000_000_000000000000_00_00000_0	P_PASS_AB (默认)	选择 A, B
%0000_0000_001_000000000000_00_00000_0	P_AND_AB	选择 A 和 B，B
%0000_0000_010_000000000000_00_00000_0	P_OR_AB	选择 A | B，B
%0000_0000_011_000000000000_00_00000_0	P_XOR_AB	选择 A ^ B，B
%0000_0000_100_000000000000_00_00000_0	FILT0_AB	选择 A, B 的 FILT0 设置
%0000_0000_101_000000000000_00_00000_0	FILT1_AB	选择 A, B 的 FILT1 设置
%0000_0000_110_000000000000_00_00000_0	FILT2_AB	选择 A, B 的 FILT2 设置
%0000_0000_111_000000000000_00_00000_0	FILT3_AB	选择 A, B 的 FILT3 设置
低级引脚模式	**打包数据模式**	
逻辑 / 施密特 / 比较器 输入模式		
%0000_0000_000_000000000000_00_00000_0	P_LOGIC_A (默认)	逻辑电平 A → 输入，输出 OUT
%0000_0000_000_000100000000_00_00000_0	P_LOGIC_A_ 反馈	逻辑电平 A → IN，输出反馈
%0000_0000_000_001000000000_00_00000_0	P_LOGIC_B_ 反馈	逻辑电平 B → 输入，输出反馈
%0000_0000_000_001100000000_00_00000_0	P_SCHMITT_A	施密特触发器 A → 输入，输出 OUT
%0000_0000_000_010000000000_00_00000_0	P_SCHMITT_A_FB	施密特触发器 A → 输入，输出反馈
%0000_0000_000_010100000000_00_00000_0	施密特触发器 B	施密特触发器 B → 输入，输出反馈
%0000_0000_000_011000000000_00_00000_0	比较 A 和 B	A B 输入，输出 OUT
%0000_0000_000_011100000000_00_00000_0	比较 A 和 B 反馈	A B 输入，输出反馈
%xxxx_xxxx_xxx_xxxxSIOHHHLLL_xx_xxxxx_x		同步模式，输入 / 输出极性，高 / 低驱动
模数转换器输入模式		
%0000_0000_000_100000000000_00_00000_0	P_ADC_GIO	模数转换器输入，输出
%0000_0000_000_100001000000_00_00000_0	P_ADC_VIO	模数转换器输入 IN，输出 OUT

Built-In Symbols for Smart Pin Configuration

Smart Pin Symbol Value	Symbol Name	Details
A Input Polarity	**(pick one)**	
%0000_0000_000_0000000000000_00_00000_0	P_TRUE_A (default)	True A input
%1000_0000_000_0000000000000_00_00000_0	P_INVERT_A	Invert A input
A Input Selection	**(pick one)**	
%0000_0000_000_0000000000000_00_00000_0	P_LOCAL_A (default)	Select local pin for A input
%0001_0000_000_0000000000000_00_00000_0	P_PLUS1_A	Select pin+1 for A input
%0010_0000_000_0000000000000_00_00000_0	P_PLUS2_A	Select pin+2 for A input
%0011_0000_000_0000000000000_00_00000_0	P_PLUS3_A	Select pin+3 for A input
%0100_0000_000_0000000000000_00_00000_0	P_OUTBIT_A	Select OUT bit for A input
%0101_0000_000_0000000000000_00_00000_0	P_MINUS3_A	Select pin-3 for A input
%0110_0000_000_0000000000000_00_00000_0	P_MINUS2_A	Select pin-2 for A input
%0111_0000_000_0000000000000_00_00000_0	P_MINUS1_A	Select pin-1 for A input
B Input Polarity	**(pick one)**	
%0000_0000_000_0000000000000_00_00000_0	P_TRUE_B (default)	True B input
%0000_1000_000_0000000000000_00_00000_0	P_INVERT_B	Invert B input
B Input Selection	**(pick one)**	
%0000_0000_000_0000000000000_00_00000_0	P_LOCAL_B (default)	Select local pin for B input
%0000_0001_000_0000000000000_00_00000_0	P_PLUS1_B	Select pin+1 for B input
%0000_0010_000_0000000000000_00_00000_0	P_PLUS2_B	Select pin+2 for B input
%0000_0011_000_0000000000000_00_00000_0	P_PLUS3_B	Select pin+3 for B input
%0000_0100_000_0000000000000_00_00000_0	P_OUTBIT_B	Select OUT bit for B input
%0000_0101_000_0000000000000_00_00000_0	P_MINUS3_B	Select pin-3 for B input
%0000_0110_000_0000000000000_00_00000_0	P_MINUS2_B	Select pin-2 for B input
%0000_0111_000_0000000000000_00_00000_0	P_MINUS1_B	Select pin-1 for B input
A, B Input Logic	**(pick one)**	
%0000_0000_000_0000000000000_00_00000_0	P_PASS_AB (default)	Select A, B
%0000_0000_001_0000000000000_00_00000_0	P_AND_AB	Select A & B, B
%0000_0000_010_0000000000000_00_00000_0	P_OR_AB	Select A | B, B
%0000_0000_011_0000000000000_00_00000_0	P_XOR_AB	Select A ^ B, B
%0000_0000_100_0000000000000_00_00000_0	P_FILT0_AB	Select FILT0 settings for A, B
%0000_0000_101_0000000000000_00_00000_0	P_FILT1_AB	Select FILT1 settings for A, B
%0000_0000_110_0000000000000_00_00000_0	P_FILT2_AB	Select FILT2 settings for A, B
%0000_0000_111_0000000000000_00_00000_0	P_FILT3_AB	Select FILT3 settings for A, B
Low-Level Pin Modes	**(pick one)**	
Logic/Schmitt/Comparator Input Modes		
%0000_0000_000_0000000000000_00_00000_0	P_LOGIC_A (default)	Logic level A → IN, output OUT
%0000_0000_000_0001000000000_00_00000_0	P_LOGIC_A_FB	Logic level A → IN, output feedback
%0000_0000_000_0010000000000_00_00000_0	P_LOGIC_B_FB	Logic level B → IN, output feedback
%0000_0000_000_0011000000000_00_00000_0	P_SCHMITT_A	Schmitt trigger A → IN, output OUT
%0000_0000_000_0100000000000_00_00000_0	P_SCHMITT_A_FB	Schmitt trigger A → IN, output feedback
%0000_0000_000_0101000000000_00_00000_0	P_SCHMITT_B_FB	Schmitt trigger B → IN, output feedback
%0000_0000_000_0110000000000_00_00000_0	P_COMPARE_AB	A > B → IN, output OUT
%0000_0000_000_0111000000000_00_00000_0	P_COMPARE_AB_FB	A > B → IN, output feedback
%xxxx_xxxx_xxx_xxxxSIOHHHLLL_xx_xxxxx_x		Sync mode, IN/output polarity, high/low drive
ADC Input Modes		
%0000_0000_000_1000000000000_00_00000_0	P_ADC_GIO	ADC GIO → IN, output OUT
%0000_0000_000_1000010000000_00_00000_0	P_ADC_VIO	ADC VIO → IN, output OUT

%0000_0000_000_1000100000000_00_00000_0	P_ADC_FLOAT	ADC FLOAT → IN, output OUT
%0000_0000_000_1000110000000_00_00000_0	P_ADC_1X	ADC 1x → IN, output OUT
%0000_0000_000_1001000000000_00_00000_0	P_ADC_3X	ADC 3.16x → IN, output OUT
%0000_0000_000_1001010000000_00_00000_0	内部时钟模数转换器 (nèibù shízhōng módù zhuǎnhuàn qì)	模数转换器 10x → 输入，输出 输出
%0000_0000_000_1001100000000_00_00000_0	内部时钟模数转换器 (nèibù shízhōng módù zhuǎnhuàn qì)	模数转换器 31.6x → 输入，输出 输出
%0000_0000_000_1001110000000_00_00000_0	内部时钟模数转换器 (nèibù shízhōng módù zhuǎnhuàn qì)	模数转换器 100x → 输入，输出 输出
%xxxx_xxxx_xxx_xxxxxxOHHHLLL_xx_xxxxx_x		输出极性，高 / 低驱动
数模转换器输出模式		**DIR 启用输出，OUT 启用模数转换器**
%0000_0000_000_1010000000000_00_00000_0	P_DAC_990R_3V	数模转换器 990Ω，3.3V 峰值，模数转换器 1x → 输入
%0000_0000_000_1010100000000_00_00000_0	P_DAC_600R_2V	数模转换器 600Ω，2.0V 峰值，模数转换器 1x → 输入
%0000_0000_000_1011000000000_00_00000_0	P_DAC_124R_3V	数模转换器 123.75Ω, 3.3V 峰值 , ADC 1x → 输入
%0000_0000_000_1011100000000_00_00000_0	P_DAC_75R_2V	数模转换器 75Ω，2.0V 峰值，模数转换器 1x → 输入
%xxxx_xxxx_xxx_xxxxxDDDDDDDD_xx_xxxxx_x		8 位 DAC 值
级比较模式		**DIR 启用输出 (1.5kΩ drive)**
%0000_0000_000_1100000000000_00_00000_0	P_ 级 A	电平输入，输出输出
%0000_0000_000_1101000000000_00_00000_0	P_ 级 A_ 负反馈	A 电平 输入，输出负反馈
%0000_0000_000_1110000000000_00_00000_0	P_ 级 B_ 正反馈	电平输入，输出正反馈
%0000_0000_000_1111000000000_00_00000_0	P_LEVEL_B_FBN	电平输入，输出负反馈
%xxxx_xxxx_xxx_xxxxSLLLLLLLL_xx_xxxxx_x		同步的，LLLLLL 电平
低电平引脚模式		
同步模式	打包数据模式	(对于逻辑 / 施密特 / 比较器 / 电平模式)
%xxxx_xxxx_xxx_xxxxSxxxxxx xx_xx_xxxxx_x		同步模式位
%0000_0000_000_00000000000_00_00000_0	P_ 异步 IO (默认)	选择 异步 IO
%0000_0000_000_0000100000000_00_00000_0	同步 I/O	选择同步 I/O
输入极性	打包数据模式	(对于逻辑 / 施密特 / 比较器模式)
%xxxx_xxxx_xxx_xxxxxIxxxxxxxx_xx_xxxxx_x		输入极性位
%0000_0000_000_00000000000_00_00000_0	真输入 (默认)	真输入位
%0000_0000_000_0000010000000_00_00000_0	反相输入	反相输入位
输出极性	打包数据模式	(用于逻辑 / 施密特 /Co 模式
%xxxx_xxxx_xxx_xxxxxxOxxxxxx_xx_xxxxx_x		输出极性位
%0000_0000_000_00000000000_00_00000_0	P_TRUE_OUTPUT (默认)	选择真输出
%0000_0000_000_0000001000000_00_00000_0	反转输出	选择反相输出
高驱动强度	打包数据模式	(对于逻辑 / 施密特 / 比较器 / 模数转换器模式)
%xxxx_xxxx_xxx_xxxxxxxHHHxxx_xx_xxxxx_x		高驱动选择位
%0000_0000_000_00000000000_00_00000_0	P_HIGH_FAST (默认)	驱动高电平快 (30mA)
%0000_0000_000_0000000001000_00_00000_0	P_HIGH_1K5	驱动高电平 1.5kΩ
%0000_0000_000_0000000010000_00_00000_0	P_HIGH_15K	驱动高电平 15kΩ
%0000_0000_000_0000000011000_00_00000_0	P_HIGH_150K	驱动高电平 150kΩ
%0000_0000_000_0000000100000_00_00000_0	P_HIGH_1mA	驱动高电平 1mA
%0000_0000_000_0000000101000_00_00000_0	P_HIGH_100UA	驱动高电平 A
%0000_0000_000_0000000110000_00_00000_0	P_HIGH_10UA	驱动高电平 A
%0000_0000_000_0000000111000_00_00000_0	高浮点	高浮点
低驱动强度	打包数据模式	(对于逻辑 / 施密特 / 比较器 / 模数转换器模式)
%xxxx_xxxx_xxx_xxxxxxxxxxLLL_xx_xxxxx_x		低驱动选择位
%0000_0000_000_00000000000_00_00000_0	低速 (默认)	低速驱动 (30mA)
%0000_0000_000_0000000000001_00_00000_0	低 1.5kΩ	降低驱动 1.5kΩ
%0000_0000_000_0000000000010_00_00000_0	低	降低驱动 15kΩ
%0000_0000_000_0000000000011_00_00000_0	低	降低驱动 150kΩ
%0000_0000_000_0000000000100_00_00000_0	低	降低驱动 1mA
%0000_0000_000_0000000000101_00_00000_0	低	降低驱动低 A
%0000_0000_000_0000000000110_00_00000_0	低	降低驱动低 A

%0000_0000_000_1000100000000_00_00000_0	P_ADC_FLOAT	ADC FLOAT → IN, output OUT
%0000_0000_000_1000110000000_00_00000_0	P_ADC_1X	ADC 1x → IN, output OUT
%0000_0000_000_1001000000000_00_00000_0	P_ADC_3X	ADC 3.16x → IN, output OUT
%0000_0000_000_1001010000000_00_00000_0	P_ADC_10X	ADC 10x → IN, output OUT
%0000_0000_000_1001100000000_00_00000_0	P_ADC_30X	ADC 31.6x → IN, output OUT
%0000_0000_000_1001110000000_00_00000_0	P_ADC_100X	ADC 100x → IN, output OUT
%xxxx_xxxx_xxx_xxxxxxOHHHLLL_xx_xxxxx_x		O = output polarity, HHH/LLL = high/low drive
DAC Output Modes		**DIR enables output, OUT enables ADC**
%0000_0000_000_1010000000000_00_00000_0	P_DAC_990R_3V	DAC 990Ω, 3.3V peak, ADC 1x → IN
%0000_0000_000_1010100000000_00_00000_0	P_DAC_600R_2V	DAC 600Ω, 2.0V peak, ADC 1x → IN
%0000_0000_000_1011000000000_00_00000_0	P_DAC_124R_3V	DAC 123.75Ω, 3.3V peak, ADC 1x → IN
%0000_0000_000_1011100000000_00_00000_0	P_DAC_75R_2V	DAC 75Ω, 2.0V peak, ADC 1x → IN
%xxxx_xxxx_xxx_xxxxxDDDDDDDD_xx_xxxxx_x		DDDDDDDD = 8-bit DAC value
Level-Comparison Modes		**DIR enables output (1.5kΩ drive)**
%0000_0000_000_1100000000000_00_00000_0	P_LEVEL_A	A > Level → IN, output OUT
%0000_0000_000_1101000000000_00_00000_0	P_LEVEL_A_FBN	A > Level → IN, output negative feedback
%0000_0000_000_1110000000000_00_00000_0	P_LEVEL_B_FBP	B > Level → IN, output positive feedback
%0000_0000_000_1111000000000_00_00000_0	P_LEVEL_B_FBN	B > Level → IN, output negative feedback
%xxxx_xxxx_xxx_xxxxSLLLLLLLL_xx_xxxxx_x		S = Synchronous, LLLLLLLL = 8-bit Level
Low-Level Pin Sub-Modes		
Sync Mode	(pick one)	(for Logic/Schmitt/Comparator/Level modes)
%xxxx_xxxx_xxx_xxxxSxxxxxxxx_xx_xxxxx_x		Sync mode bit
%0000_0000_000_0000000000000_00_00000_0	P_ASYNC_IO (default)	Select asynchronous I/O
%0000_0000_000_0000100000000_00_00000_0	P_SYNC_IO	Select synchronous I/O
IN Polarity	(pick one)	(for Logic/Schmitt/Comparator modes)
%xxxx_xxxx_xxx_xxxxxIxxxxxxxx_xx_xxxxx_x		IN polarity bit
%0000_0000_000_0000000000000_00_00000_0	P_TRUE_IN (default)	True IN bit
%0000_0000_000_0000010000000_00_00000_0	P_INVERT_IN	Invert IN bit
Output Polarity	(pick one)	(for Logic/Schmitt/Comparator/ADC modes)
%xxxx_xxxx_xxx_xxxxxxOxxxxxx_xx_xxxxx_x		Output polarity bit
%0000_0000_000_0000000000000_00_00000_0	P_TRUE_OUTPUT (default)	Select true output
%0000_0000_000_0000001000000_00_00000_0	P_INVERT_OUTPUT	Select inverted output
Drive-High Strength	(pick one)	(for Logic/Schmitt/Comparator/ADC modes)
%xxxx_xxxx_xxx_xxxxxxxHHHxxx_xx_xxxxx_x		Drive-high selector bits
%0000_0000_000_0000000000000_00_00000_0	P_HIGH_FAST (default)	Drive high fast (30mA)
%0000_0000_000_0000000001000_00_00000_0	P_HIGH_1K5	Drive high 1.5kΩ
%0000_0000_000_0000000010000_00_00000_0	P_HIGH_15K	Drive high 15kΩ
%0000_0000_000_0000000011000_00_00000_0	P_HIGH_150K	Drive high 150kΩ
%0000_0000_000_0000000100000_00_00000_0	P_HIGH_1MA	Drive high 1mA
%0000_0000_000_0000000101000_00_00000_0	P_HIGH_100UA	Drive high 100µA
%0000_0000_000_0000000110000_00_00000_0	P_HIGH_10UA	Drive high 10µA
%0000_0000_000_0000000111000_00_00000_0	P_HIGH_FLOAT	Float high
Drive-Low Strength	(pick one)	(for Logic/Schmitt/Comparator/ADC modes)
%xxxx_xxxx_xxx_xxxxxxxxxxLLL_xx_xxxxx_x		Drive-low selector bits
%0000_0000_000_0000000000000_00_00000_0	P_LOW_FAST (default)	Drive low fast (30mA)
%0000_0000_000_0000000000001_00_00000_0	P_LOW_1K5	Drive low 1.5kΩ
%0000_0000_000_0000000000010_00_00000_0	P_LOW_15K	Drive low 15kΩ
%0000_0000_000_0000000000011_00_00000_0	P_LOW_150K	Drive low 150kΩ
%0000_0000_000_0000000000100_00_00000_0	P_LOW_1MA	Drive low 1mA
%0000_0000_000_0000000000101_00_00000_0	P_LOW_100UA	Drive low 100µA
%0000_0000_000_0000000000110_00_00000_0	P_LOW_10UA	Drive low 10µA

%0000_0000_000_00000000000111_00_00000_0	低浮点数	浮点数 低
输出控制	**打包数据模式**	
%0000_0000_000_00000000000_00_00000_0	P_TT_00 (默认)	TT
%0000_0000_000_00000000000_01_00000_0	P_TT_01	TT =
%0000_0000_000_00000000000_10_00000_0	TT_10	TT = %10
%0000_0000_000_00000000000_11_00000_0	P_TT_11	TT = %11
%0000_0000_000_00000000000_01_00000_0	P_OE	启用智能引脚模式输出
%0000_0000_000_00000000000_01_00000_0	P_CHANNEL	在非智能别针 DAC 模式下启用 DAC 通道
%0000_0000_000_00000000000_10_00000_0	P_BITDAC	在非智能引脚 DAC 模式下启用 BITDAC
智能别针模式	**打包数据模式**	
%0000_0000_000_00000000000_00_00000_0	正常模式 (默认)	正常模式（非智能别针模式）
%0000_0000_000_00000000000_00_00001_0	P_ 仓库	长存储器 (非数模转换器模式)
%0000_0000_000_00000000000_00_00001_0	数模转换器噪声	数模转换器噪声 (DAC 模式)
%0000_0000_000_00000000000_00_00010_0	DAC 抖动随机	16 位随机抖动（ DAC 模式）
%0000_0000_000_00000000000_00_00011_0	P_DAC_ 抖动 PWM	16 位 DAC PWM 抖动 (DAC 模式)
%0000_0000_000_00000000000_00_00100_0	P_ 脉冲	脉冲 / 周期输出
%0000_0000_000_00000000000_00_00101_0	P_ 切换	切换输出
%0000_0000_000_00000000000_00_00110_0	P_ 非线性数字控制频率	非线性数字控制频率输出
%0000_0000_000_00000000000_00_00111_0	P_NCO_DUTY	NCO 职责 输出
%0000_0000_000_00000000000_00_01000_0	P_PWM_TRIANGLE	PWM 三角波 输出
%0000_0000_000_00000000000_00_01001_0	P_PWM_SAWTOOOTH	PWM 锯齿波 输出
%0000_0000_000_00000000000_00_01010_0	P_PWM_SMPS	PWM 开关电源 I/O
%0000_0000_000_00000000000_00_01011_0	P_QUADRATURE	A-B 四面切正编码输入
%0000_0000_000_00000000000_00_01100_0	P_REG_UP	当 B- 高 升高时，A 增加。
%0000_0000_000_00000000000_00_01101_0	P_REG_UP_DOWN	当 B- 高 时，A-rise 上升；当 B- 低 时，A-rise 下降。
%0000_0000_000_00000000000_00_01110_0	P_COUNT_RISES	在 A-rise 上，可选地在 B-rise 上进行解码
%0000_0000_000_00000000000_00_01111_0	P_COUNT_HIGHS	在 A- 高上，可选地在 B- 高上进行降噪。
%0000_0000_000_00000000000_00_10000_0	P_ 状态计数	对于 A- 低和 A- 高状态，计数计数。
%0000_0000_000_00000000000_00_10001_0	P_ 高计数	对于 A- 高状态，计数 计数
%0000_0000_000_00000000000_00_10010_0	P_ 事件计数	对于 X A- 高 / 上升 / 边缘，计数计数 / 在没有 A- 高 / 上升 / 边沿信号的情况下，经过 X 计数后超时
%0000_0000_000_00000000000_00_10011_0	P_ 周期计数	X 周期的 A，计数
%0000_0000_000_00000000000_00_10100_0	周期高点	计算 X 时期内 A 的高点。
%0000_0000_000_00000000000_00_10101_0	计数器计数	在 X+ 计数期间的 A 期间，计数计数。
%0000_0000_000_00000000000_00_10110_0	计数器高点	在 X+ 计数周期中，计算 A 的高点。
%0000_0000_000_00000000000_00_10111_0	计数器周期	在 X+ 计数时期内，计算 A 期间。
%0000_0000_000_00000000000_00_11000_0	内部时钟模数转换器 (nèibù shízhōng módù zhuǎnhuàn qì)	内部时钟的模数转换器采样 / 滤波 / 捕获
%0000_0000_000_00000000000_00_11001_0	外部时钟模数转换器 (wàibù shízhōng módù zhuǎnhuàn qì)	模数转换器 采样 / 滤波 / 捕获，外部时钟
%0000_0000_000_00000000000_00_11010_0	带触发的模数转换器示波器	带触发的模数转换器范围
%0000_0000_000_00000000000_00_11011_0	USB 引脚对 (USB yǐnjiǎo duì)	USB 针脚对
%0000_0000_000_00000000000_00_11100_0	同步串行发送	同步的串行发送
%0000_0000_000_00000000000_00_11101_0	同步串行接收	同步的串行接收 |
%0000_0000_000_00000000000_00_11110_0	异步串行发送	异步的串行发送
%0000_0000_000_00000000000_00_11111_0	异步串行接收	异步的串行接收

%0000_0000_000_00000000000111_00_00000_0	P_LOW_FLOAT	Float low
DIR/OUT Control	**(pick one)**	
%0000_0000_000_00000000000000_00_00000_0	P_TT_00 (default)	TT = %00
%0000_0000_000_00000000000000_01_00000_0	P_TT_01	TT = %01
%0000_0000_000_00000000000000_10_00000_0	P_TT_10	TT = %10
%0000_0000_000_00000000000000_11_00000_0	P_TT_11	TT = %11
%0000_0000_000_00000000000000_01_00000_0	P_OE	Enable output in smart pin mode
%0000_0000_000_00000000000000_01_00000_0	P_CHANNEL	Enable DAC channel in non-smart pin DAC mode
%0000_0000_000_00000000000000_10_00000_0	P_BITDAC	Enable BITDAC for non-smart pin DAC mode
Smart Pin Modes	**(pick one)**	
%0000_0000_000_00000000000000_00_00000_0	P_NORMAL (default)	Normal mode (not smart pin mode)
%0000_0000_000_00000000000000_00_00001_0	P_REPOSITORY	Long repository (non-DAC mode)
%0000_0000_000_00000000000000_00_00001_0	P_DAC_NOISE	DAC Noise (DAC mode)
%0000_0000_000_00000000000000_00_00010_0	P_DAC_DITHER_RND	DAC 16-bit random dither (DAC mode)
%0000_0000_000_00000000000000_00_00011_0	P_DAC_DITHER_PWM	DAC 16-bit PWM dither (DAC mode)
%0000_0000_000_00000000000000_00_00100_0	P_PULSE	Pulse/cycle output
%0000_0000_000_00000000000000_00_00101_0	P_TRANSITION	Transition output
%0000_0000_000_00000000000000_00_00110_0	P_NCO_FREQ	NCO frequency output
%0000_0000_000_00000000000000_00_00111_0	P_NCO_DUTY	NCO duty output
%0000_0000_000_00000000000000_00_01000_0	P_PWM_TRIANGLE	PWM triangle output
%0000_0000_000_00000000000000_00_01001_0	P_PWM_SAWTOOTH	PWM sawtooth output
%0000_0000_000_00000000000000_00_01010_0	P_PWM_SMPS	PWM switch-mode power supply I/O
%0000_0000_000_00000000000000_00_01011_0	P_QUADRATURE	A-B quadrature encoder input
%0000_0000_000_00000000000000_00_01100_0	P_REG_UP	Inc on A-rise when B-high
%0000_0000_000_00000000000000_00_01101_0	P_REG_UP_DOWN	Inc on A-rise when B-high, dec on A-rise when B-low
%0000_0000_000_00000000000000_00_01110_0	P_COUNT_RISES	Inc on A-rise, optionally dec on B-rise
%0000_0000_000_00000000000000_00_01111_0	P_COUNT_HIGHS	Inc on A-high, optionally dec on B-high
%0000_0000_000_00000000000000_00_10000_0	P_STATE_TICKS	For A-low and A-high states, count ticks
%0000_0000_000_00000000000000_00_10001_0	P_HIGH_TICKS	For A-high states, count ticks
%0000_0000_000_00000000000000_00_10010_0	P_EVENTS_TICKS	For X A-highs/rises/edges, count ticks / Timeout on X ticks of no A-high/rise/edge
%0000_0000_000_00000000000000_00_10011_0	P_PERIODS_TICKS	For X periods of A, count ticks
%0000_0000_000_00000000000000_00_10100_0	P_PERIODS_HIGHS	For X periods of A, count highs
%0000_0000_000_00000000000000_00_10101_0	P_COUNTER_TICKS	For periods of A in X+ ticks, count ticks
%0000_0000_000_00000000000000_00_10110_0	P_COUNTER_HIGHS	For periods of A in X+ ticks, count highs
%0000_0000_000_00000000000000_00_10111_0	P_COUNTER_PERIODS	For periods of A in X+ ticks, count periods
%0000_0000_000_00000000000000_00_11000_0	P_ADC	ADC sample/filter/capture, internally clocked
%0000_0000_000_00000000000000_00_11001_0	P_ADC_EXT	ADC sample/filter/capture, externally clocked
%0000_0000_000_00000000000000_00_11010_0	P_ADC_SCOPE	ADC scope with trigger
%0000_0000_000_00000000000000_00_11011_0	P_USB_PAIR	USB pin pair
%0000_0000_000_00000000000000_00_11100_0	P_SYNC_TX	Synchronous serial transmit
%0000_0000_000_00000000000000_00_11101_0	P_SYNC_RX	Synchronous serial receive
%0000_0000_000_00000000000000_00_11110_0	P_ASYNC_TX	Asynchronous serial transmit
%0000_0000_000_00000000000000_00_11111_0	P_ASYNC_RX	Asynchronous serial receive

流媒体模式的内置符号

符号值	符号名称
即时 → LUT → 引脚 / 数模转换器	
%0000_0000_0000_0000 << 16 %0000_DDDD_EPPP_BBBB << 16	X_IMM_32X1_LUT
%0001_0000_0000_0000 << 16 %0001_DDDD_EPPP_BBBB << 16	X_IMM_16X2_LUT
%0010_0000_0000_0000 << 16 %0010_DDDD_EPPP_BBBB << 16	X_IMM_8X4_LUT
%0011_0000_0000_0000 << 16 %0011_DDDD_EPPP_BBBB << 16	X_IMM_4X8_LUT
立即引脚 / 数模转换器	
%0100_0000_0000_0000 << 16 %0100_DDDD_EPPP_PPPA << 16	X_IMM_32X1_1DAC1
%0101_0000_0000_0000 << 16 %0101_DDDD_EPPP_PP0A << 16	X_IMM_16X2_2DAC1
%0101_0000_0000_0010 << 16 %0101_DDDD_EPPP_PP1A << 16	X_IMM_16X2_1DAC2
%0110_0000_0000_0000 << 16 %0110_DDDD_EPPP_P00A << 16	X_IMM_8X4_4DAC1
%0110_0000_0000_0010 << 16 %0110_DDDD_EPPP_P01A << 16	X_IMM_8X4_2DAC2
%0110_0000_0000_0100 << 16 %0110_DDDD_EPPP_P10A << 16	X_IMM_8X4_1DAC4
%0110_0000_0000_0110 << 16 %0110_DDDD_EPPP_0110 << 16	X_IMM_4X8_4DAC2
%0110_0000_0000_0111 << 16 %0110_DDDD_EPPP_0111 << 16	X_IMM_4X8_2DAC4
%0110_0000_0000_1110 << 16 %0110_DDDD_EPPP_1110 << 16	X_IMM_4X8_1DAC8
%0110_0000_0000_1111 << 16 %0110_DDDD_EPPP_1111 << 16	X_IMM_2X16_4DAC4
%0111_0000_0000_0000 << 16 %0111_DDDD_EPPP_0000 << 16	X_IMM_2X16_2DAC8
%0111_0000_0000_0001 << 16 %0111_DDDD_EPPP_0001 << 16	X_IMM_1X32_4DAC8
RDFAST → LUT → 引脚 / 数模转换器	
%0111_0000_0000_0010 << 16 %0111_DDDD_EPPP_001A << 16	X_RFLONG_32X1_LUT
%0111_0000_0000_0100 << 16 %0111_DDDD_EPPP_010A << 16	X_RFLONG_16X2_LUT
%0111_0000_0000_0110 << 16 %0111_DDDD_EPPP_011A << 16	X_RFLONG_8X4_LUT
%0111_0000_0000_1000 << 16 %0111_DDDD_EPPP_1000 << 16	X_RFLONG_4X8_LUT
RDFAST 引脚 / 数模转换器	
%1000_0000_0000_0000 << 16 %1000_DDDD_EPPP_PPPA << 16	X_RFBYTE_1P_1DAC1
%1001_0000_0000_0000 << 16 %1001_DDDD_EPPP_PP0A << 16	X_RFBYTE_2P_2DAC1
%1001_0000_0000_0010 << 16 %1001_DDDD_EPPP_PP1A << 16	X_RFBYTE_2P_1DAC2
%1010_0000_0000_0000 << 16 %1010_DDDD_EPPP_P00A << 16	X_RFBYTE_4P_4DAC1
%1010_0000_0000_0010 << 16 %1010_DDDD_EPPP_P01A << 16	X_RFBYTE_4P_2DAC2
%1010_0000_0000_0100 << 16 %1010_DDDD_EPPP_P10A << 16	X_RFBYTE_4P_1DAC4
%1010_0000_0000_0110 << 16 %1010_DDDD_EPPP_0110 << 16	X_RFBYTE_8P_4DAC2
%1010_0000_0000_0111 << 16 %1010_DDDD_EPPP_0111 << 16	X_RFBYTE_8P_2DAC4
%1010_0000_0000_1110 << 16 %1010_DDDD_EPPP_1110 << 16	X_RFBYT_8P_1DAC8
%1010_0000_0000_1111 << 16	X_RFWORD_16P_4DAC4

Built-In Symbols for Streamer Modes

Streamer Symbol Value	Symbol Name
#Immediate → LUT → Pins / DACs	
%0000_0000_0000_0000 << 16 %0000_DDDD_EPPP_BBBB << 16	X_IMM_32X1_LUT
%0001_0000_0000_0000 << 16 %0001_DDDD_EPPP_BBBB << 16	X_IMM_16X2_LUT
%0010_0000_0000_0000 << 16 %0010_DDDD_EPPP_BBBB << 16	X_IMM_8X4_LUT
%0011_0000_0000_0000 << 16 %0011_DDDD_EPPP_BBBB << 16	X_IMM_4X8_LUT
#Immediate → Pins / DACs	
%0100_0000_0000_0000 << 16 %0100_DDDD_EPPP_PPPA << 16	X_IMM_32X1_1DAC1
%0101_0000_0000_0000 << 16 %0101_DDDD_EPPP_PP0A << 16	X_IMM_16X2_2DAC1
%0101_0000_0000_0010 << 16 %0101_DDDD_EPPP_PP1A << 16	X_IMM_16X2_1DAC2
%0110_0000_0000_0000 << 16 %0110_DDDD_EPPP_P00A << 16	X_IMM_8X4_4DAC1
%0110_0000_0000_0010 << 16 %0110_DDDD_EPPP_P01A << 16	X_IMM_8X4_2DAC2
%0110_0000_0000_0100 << 16 %0110_DDDD_EPPP_P10A << 16	X_IMM_8X4_1DAC4
%0110_0000_0000_0110 << 16 %0110_DDDD_EPPP_0110 << 16	X_IMM_4X8_4DAC2
%0110_0000_0000_0111 << 16 %0110_DDDD_EPPP_0111 << 16	X_IMM_4X8_2DAC4
%0110_0000_0000_1110 << 16 %0110_DDDD_EPPP_1110 << 16	X_IMM_4X8_1DAC8
%0110_0000_0000_1111 << 16 %0110_DDDD_EPPP_1111 << 16	X_IMM_2X16_4DAC4
%0111_0000_0000_0000 << 16 %0111_DDDD_EPPP_0000 << 16	X_IMM_2X16_2DAC8
%0111_0000_0000_0001 << 16 %0111_DDDD_EPPP_0001 << 16	X_IMM_1X32_4DAC8
RDFAST → LUT → Pins / DACs	
%0111_0000_0000_0010 << 16 %0111_DDDD_EPPP_001A << 16	X_RFLONG_32X1_LUT
%0111_0000_0000_0100 << 16 %0111_DDDD_EPPP_010A << 16	X_RFLONG_16X2_LUT
%0111_0000_0000_0110 << 16 %0111_DDDD_EPPP_011A << 16	X_RFLONG_8X4_LUT
%0111_0000_0000_1000 << 16 %0111_DDDD_EPPP_1000 << 16	X_RFLONG_4X8_LUT
RDFAST → Pins / DACs	
%1000_0000_0000_0000 << 16 %1000_DDDD_EPPP_PPPA << 16	X_RFBYTE_1P_1DAC1
%1001_0000_0000_0000 << 16 %1001_DDDD_EPPP_PP0A << 16	X_RFBYTE_2P_2DAC1
%1001_0000_0000_0010 << 16 %1001_DDDD_EPPP_PP1A << 16	X_RFBYTE_2P_1DAC2
%1010_0000_0000_0000 << 16 %1010_DDDD_EPPP_P00A << 16	X_RFBYTE_4P_4DAC1
%1010_0000_0000_0010 << 16 %1010_DDDD_EPPP_P01A << 16	X_RFBYTE_4P_2DAC2
%1010_0000_0000_0100 << 16 %1010_DDDD_EPPP_P10A << 16	X_RFBYTE_4P_1DAC4
%1010_0000_0000_0110 << 16 %1010_DDDD_EPPP_0110 << 16	X_RFBYTE_8P_4DAC2
%1010_0000_0000_0111 << 16 %1010_DDDD_EPPP_0111 << 16	X_RFBYTE_8P_2DAC4
%1010_0000_0000_1110 << 16 %1010_DDDD_EPPP_1110 << 16	X_RFBYTE_8P_1DAC8
%1010_0000_0000_1111 << 16	X_RFWORD_16P_4DAC4